B1.1 Definition of AXI4-Lite

This section defines the functionality and signal requirements of AXI4-Lite components.

The key functionality of AXI4-Lite operation is:

- all transactions are of burst length 1
- all data accesses use the full width of the data bus
  - AXI4-Lite supports a data bus width of 32-bit or 64-bit.
- all accesses are Non-modifiable, Non-bufferable
- Exclusive accesses are not supported.

B1.1.1 Signal list

Table B1-1 shows the required signals on an AXI4-Lite interface.

Table B1-1 AXI4-Lite interface signals

Global	Write address channel	Write data channel	Write response channel	Read address channel	Read data channel
ACLK	AWVALID	WVALID	BVALID	ARVALID	RVALID
ARESETn	AWREADY	WREADY	BREADY	ARREADY	RREADY
–	AWADDR	WDATA	BRESP	ARADDR	RDATA
–	AWPROT	WSTRB	–	ARPROT	RRESP

AXI4 signals modified in AXI4-Lite

The AXI4-Lite interface does not fully support the following signals:

RRESP, BRESP

The EXOKAY response is not supported on the read data and write response channels.

AXI4 signals not supported in AXI4-Lite

The AXI4-Lite interface does not support the following signals:

AWLEN, ARLEN The burst length is defined to be 1, equivalent to an **AxLEN** value of zero.

AWSIZE, ARSIZE All accesses are defined to be the width of the data bus.

Note

AXI4-Lite requires a fixed data bus width of either 32-bits or 64-bits.

AWBURST, ARBURST

The burst type has no meaning because the burst length is 1.

AWLOCK, ARLOCK

All accesses are defined as Normal accesses, equivalent to an **AxLOCK** value of zero.

AWCACHE, ARCACHE

All accesses are defined as Non-modifiable, Non-bufferable, equivalent to an **AxCACHE** value of `0b0000`.

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WLAST, RLAST All bursts are defined to be of length 1, equivalent to a **WLAST** or **RLAST** value of 1.

B1.1.2 Bus width

AXI4-Lite has a fixed data bus width and all transactions are the same width as the data bus. The data bus width must be, either 32-bits or 64-bits.

ARM expects that:

- the majority of components use a 32-bit interface
- only components requiring 64-bit atomic accesses use a 64-bit interface.

A 64-bit component can be designed for access by 32-bit masters, but the implementation must ensure that the component sees all transactions as 64-bit transactions.

Note

This interoperability can be achieved by including, in the register map of the component, locations that are suitable for access by a 32-bit master. Typically, such locations would use only the lower 32-bits of the data bus.

B1.1.3 Write strobes

The AXI4-Lite protocol supports write strobes. This means multi-sized registers can be implemented and also supports memory structures that require support for 8-bit and 16-bit accesses.

All master interfaces and interconnect components must provide correct write strobes.

Any slave component can choose whether to use the write strobes. The options permitted are:

- to make full use of the write strobes
- to ignore the write strobes and treat all write accesses as being the full data bus width
- to detect write strobe combinations that are not supported and provide an error response.

A slave that provides memory access must fully support write strobes. Other slaves in the memory map might support a more limited write strobe option.

When converting from full AXI to AXI4-Lite, a write transaction can be generated on AXI4-Lite with all write strobes deasserted. Automatic suppression of such transactions is permitted but not required. See *Conversion, protection, and detection* on page B1-127.

B1.1.4 Optional signaling

AXI4-Lite supports multiple outstanding transactions, but a slave can restrict this by the appropriate use of the handshake signals.

AXI4-Lite does not support AXI IDs. This means all transactions must be in order, and all accesses use a single fixed ID value.

Note

Optionally, an AXI4-Lite slave can support AXI ID signals, so that it can be connected to a full AXI interface without modification. See *Interoperability* on page B1-124.

AXI4-Lite does not support data interleaving, the burst length is defined as 1.

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